THE ROLE OF COMPUTED TOMOGRAPHY IN ADDITIVE MANUFACTURING

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INTRODUCTION

Additive manufacturing (AM) technologies—or 3D printing, as they are popularly known—show promise as a way to transform traditional production manufacturing because they can produce highly complex geometries and customized parts directly from the part design model without dedicated tooling. However, many questions about the structural integrity of 3D printed parts—tolerance limits, layer defects, residual stress, and material inclusions—remain unanswered because AM process parameters and disruptions during the material layering (generally in powder form) may induce a variety of dimensional deviations and internal flaws (e.g., cracks or voids) in the final product. These flaws might affect the performance of AM devices and create risks of potential failures, so the support of metrology and non-destructive testing (NDT) techniques for better assessment of AM parts is needed [1-5].

One of the challenges to the assessment of AM parts is that many will have internal features, these are generally inaccessible from the outside to vision and contact-based inspection techniques for quality control. While destructive methods can be used to extract measurements, dimensional information from the disassembled state of a product may differ from the actual geometrical dimensions in the original assembled state. This paper describes some of the main issues associated with the measurement of AM parts and some future trends for the development of alternative techniques for measuring complex shaped AM parts. Along with a brief discussion of limitations of traditional inspection technologies, such as coordinate measuring machines (CMMs) and optical-based systems, the particular case of X-ray computed tomography (CT) as a technology to support AM inspection and development is discussed. The benefits of X-ray CT for the assessment of the structural integrity of AM parts and deviations typically encountered in AM dimensional geometry when compared to reference/nominal geometry will be considered.

AM QUALIFICATION CHALLENGES

Due to its almost unlimited capability to produce highly complex geometries and customized parts directly from the part model, AM techniques have recently received significant attention as technologies that may have the potential to transform traditional manufacturing techniques as a more cost-effective production process. AM is being considered for safety-critical components, such as those found in propulsion engine components and customized biomedical implants. Consequently, development of AM technologies should be accompanied by quality control processes and inspection technologies. The need for established methods of AM qualification has been previously highlighted by several authors [1-7]. The rationale underpinning the call for AM qualification usually appeals to one or more of the following necessities: to understand the processing techniques and geometry impact on AM material properties; to understand and develop AM process optimization; to design methods of engineering that take advantage of the AM possibilities; to identify suitable parts for AM; to develop qualification standards for AM materials and quality control processes; and to develop in-situ measurement systems for inspecting AM processes. Whereas some researchers may stress the importance of raw material control and identifying the key variables impacting manufacture, others tend to accentuate a qualification approach that focuses on inspecting finished AM parts by combining the existing knowledge of qualification methods for conventional manufacturing processes. In reality, these approaches are related and connected because the impact of product geometry on the material microstructure is an inherent characteristic of the AM processes that, in turn, directly affects the final product. However, the qualification of finished AM parts is challenging because such parts can be complex in form (e.g. freeform-shaped parts). They typically have a high surface roughness, and they may also have internal geometry with interconnected structures and channels. As such, they can potentially contain voids and inclusions (such as un-melted particles or powder residues) inside their volume. These challenges make it difficult for conventional NDT techniques—such as ultrasonic, infrared, eddy current, radiographic inspection, and light-based technologies—to achieve comprehensive inspections on the quality of AM parts. X-ray CT is an alternative technique that proves useful for examining parts of complex internal geometry and can quantitatively identify many of the aforementioned defects. Although it still has limitations related to resolution, sensitivity, and speed, industrial CT equipment continues to develop and improve technologically. As a result, X-ray CT has been successful in addressing some of the quality inspection needs for the AM industry and is increasingly used as a tool for qualifying AM parts.

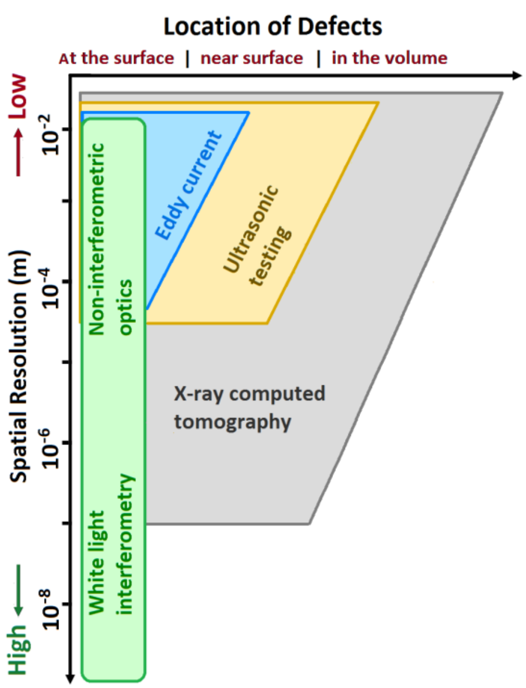

FIGURE 1. Comparison of some relevant NDT techniques according to detectable defect location and their achievable spatial resolution. Adapted from [8].

Figure 1 shows a comparison of NDT techniques typically used for defect detection in AM parts for dimensional measurements, which is categorized according to detectable defect location and spatial resolution. Accordingly, optical methods can achieve very high measuring resolutions, in the range of nanometers when interferometry techniques are employed, but they can only inspect defects at the surface (or through a surface opening). Eddy-current testing and ultrasonic techniques can detect defects within the volume if they are not located very deep inside the testing sample, but the one drawback is the limited spatial resolution of detection, which is in the millimeter range or some fraction of millimeters in the most optimal situations and for even more limited depths into the surface. In contrast, the best method for nondestructive inspection of complex structures and geometries inside the volume of a part is currently X-ray CT, with a resolution from millimeter to micrometer ranges, and sub-micron levels in some special cases, e.g., by using synchrotron X-ray sources or with the help of focusing elements such as Fresnel zone plates or Kirkpatrick-Baez mirrors [9-12].

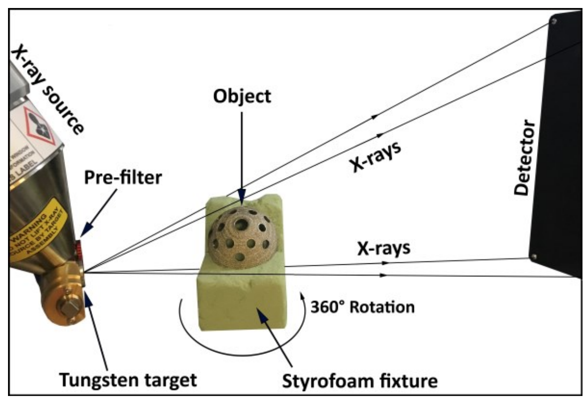

FIGURE 2. Basic setup for an X-ray CT scan. After collecting several radiographic images—also known as projection images (or simply 'projections')—taken from a large number of different angular positions around the specimen during a 360º rotation, with the aid of a reconstruction algorithm, it is possible to reconstruct a three-dimensional image of the object detailing all of its internal and external features.

X-RAY COMPUTED TOMOGRAPHY

Since CT shows several advantages as a non-destructive method for acquiring structural characterization of both internal and external geometries of AM parts, it is often the only viable option to extract component dimensions of internal or hidden features [13-16]. This makes CT technology a partner in the "3D printing revolution" for the study and inspection of AM products. Thus X-ray CT is typically used to detect porosity and cracks in AM parts, dimensional deviations from computer-aided design (CAD) models, and powder residues or inclusions left inside the AM built part. Figure 2 shows a typical CT setup featuring the scanning of an AM acetabular hip prosthesis cup made of Ti-6Al-4V and produced by an electron beam melting process, which has a prescribed non-uniform lattice structure forming struts over the surface with an interconnected porosity (to stimulate bone adhesion). In recent work [17], along with the use of image processing software such as MATLAB and ImageJ, X-ray CT has been used to provide quality inspection and characterization of the porous structure that affects the implant's fixation. In such an application, the non-destructive and non-contact nature of CT shows a clear advantage over other inspection methods, such as tactile CMMs and optical systems; it provides an analysis of interconnectivity of the porous structure, the standard deviation of the size of the pores and struts, and the local thickness of the lattice structure in its size and spatial location [17]. Additional examples of the X-ray CT usage for inspection of AM parts and processes can be found elsewhere, e.g., [18-19].

In general, tactile CMMs or optical measuring instruments like laser scanners are limited to the measurement of the external surface of an AM part and can provide additional measurements for partial qualification of CT measurements. In addition, tactile CMMs can produce compressive stresses and friction during sliding that could produce wear at the surface [20-21]. In contrast, X-ray CT eliminates the above difficulties because it is a non-contact technique that can access internal features. Since the principles of X-ray CT—when used for obtaining dimensional measurements—differ substantially from tactile CMMs and optical or light-based systems, each technique has its own distinctive attributes that accentuate and limit their capabilities and range of applications. Table 1 shows a comparative list of some of the main factors influencing the metrological performance of tactile CMMs, optical systems, and X-ray CT machines. Naturally, the capability of defect detection by X-ray CT depends very much on the part's size, complexity, and material, but in principle, high scanning metrological resolutions in the order of 10 or 5 µm are possible [22-23].

Table 1. Main influencing factors that can affect the performance of optical, tactile, and X-ray CT inspection systems for dimensional measurement tasks [21].

Optical	Tactile	X-ray CT
Resolution limit (diffraction)	Diameter of stylus tip	Resolution and magnification
Optical magnification	Stylus shaft length and stiffness	Image contrast/signal-to-noise
Sensor receptor gain/reading	Sampling strategy	Detector gain/sensitivity
Surface slope or tilt	Number of probing points	Number of projections
Surface roughness/finish	Surface characteristics	X-rays energy spectrum
Object geometry and size	Object geometry and size	Object geometry and size
Illumination type/intensity	Object clamping/mounting	Object orientation/tilt
Material transparency/translucency	Probing speed/force/direction	Material composition
Surface reflectivity/opacity	Probe calibration	Scaling calibration
Edge sub-pixel interpolation	Fitting algorithms	Edge sub-voxel interpolation
Environmental conditions	Environmental conditions	Environmental conditions

DEFECT ANALYSIS WITH X-RAY CT

Recently, there has been a growing interest in the determination of material porosity in AM manufactured parts, e.g., see [24-26]. In cases of AM processes that use metal powders, the powder typically comes in solid grains, and if such grains are fine, the powder particles are within the size range of 15 µm to 150 µm [27]. However, some of the powder particles might be hollow, i.e., some of them may come with pores inside them, as can be seen in Figure 3. In addition to porosity, particle size distribution, morphology, heterogeneity, geometry, and grain microstructure are of particular interest in AM powders [27-31]. There are certain anomalies that could potentially induce defects in powder-based AM parts. For example, powders with uniform size distribution are known to promote homogeneous melting, and good interlayer bonding [29]. In contrast, powders prone to form grain agglomerates may assume irregular shapes or lead to porosity in the component when metal AM laser or electron beam melting is used. As can be seen from the right image (in Figure 3), most grains of aluminum powder seems to have shapes that are non-spherical, of different size, and irregular morphology. Several of them are agglomerated in small clusters. In addition, several grains show porosity channels or voids in their internal structure. In contrast, as seen on the left side of Figure 3, Inconel powder grains seem to be consistently round, almost identical in size, and relatively small as compared to the aluminum powder. However, some of the Inconel powder grains seem to still exhibit micro-porosity structures inside them. The presence of these channels or microstructures in the powder material could potentially influence the porosity percentage of a finished AM part built from such material or induce fractures and cracks inside by the lack of fusion voids. In the current literature, there are several studies on the influence of powder related parameters—such as particle size, uniformity, shape and distribution, or grain porosity—on the final result of an AM process, including evidence of part defects induced by unmolten powder particles or agglomeration of powder trapped in the inside, e.g., see [28-33]. Figure 5 illustrates the application of X-ray CT for the inspection of porosity in a finished AM part, revealing a variety of voids of different size, shape, and frequency distribution.

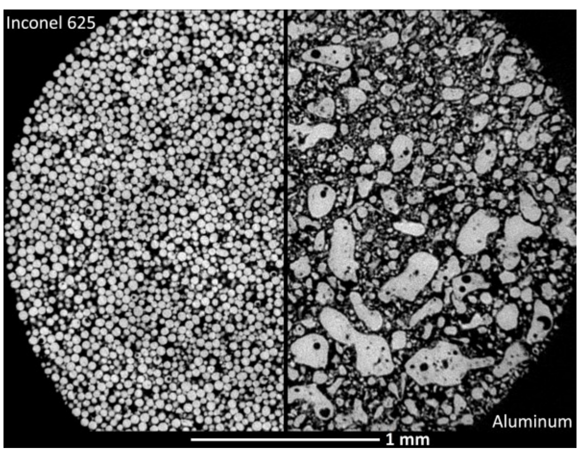

FIGURE 3. A composed X-ray CT image of two different raw AM powders that reveals porosity inside the AM particles. Left: nickel-chromium alloy (Inconel 625). Right: aluminum 225 mesh (65 µm). Data were obtained using identical CT geometrical configuration (with a Nikon XTH 225 ST system) but different X-ray power settings: 93 kV and 50 µm for the Inconel sample, and 60 kV and 95 µA for the aluminum powder.

In a broader sense, when testing and qualifying AM components, it is vital to understand defects effects on the structural performance and overall quality of the final product [4, 24-26]. For example, along with an additional AM cylindrical rod part acting as ‘witness coupon’, the turbine blade from Figure 5 has previously been used to illustrate the usefulness of CT data on the study of property/characteristics transfer from a ‘witness coupon’ to a component (made at the same time and of the same material as the actual component). Initial results of these studies, as presented elsewhere [4], show that non-fracture-critical properties may have a much lower dependency on the defect distribution in comparison to fatigue.

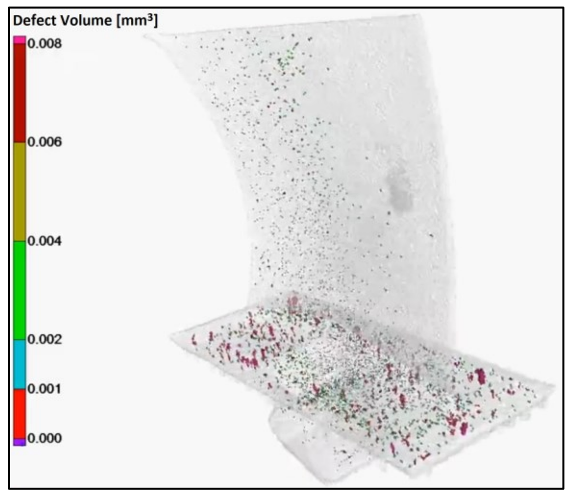

FIGURE 5. X-ray CT porosity analysis of an AM turbine blade, of approximately 5.5 cm height, made of Ti-6Al-4V, and produced by electron beam melting process.

In actuality, measuring and characterizing each internal defect from the CT data can be a daunting task, mainly due to the sheer quantity of data that resides in high-resolution, volumetric CT datasets. Fortunately, many tools are available to assist in the measurement and analysis of large quantities of internal defects, e.g., see [34-37], and automated defect recognition algorithms are increasingly being implemented in practice, including tools for image processing and statistical analysis techniques that catalogue and summarize defects inside a particular volume. In the end, whether or not to pass or fail an AM component will need to rely upon comprehensive knowledge of the defects and their relationship to material properties and input process parameters. Additional testing and more advanced analysis may be required to establish the proper baseline to determine what level of defects constitutes a failed part.

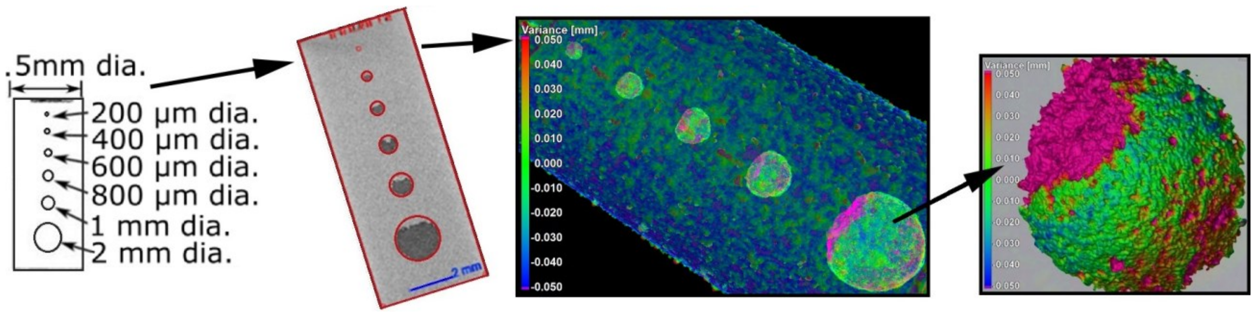

FIGURE 6. X-ray CT inspection of an AM test part (designed and fabricated by NIST) with an internal structure intended to recreate flaws inside the part. The test part was made of a nickel-chromium alloy (Inconel 625) and produced by a laser-based powder bed fusion process. The color-coded bar features dimensional deviations in the geometry of the AM part when compared to the nominal intended design, i.e., with respect to the reference/nominal geometry of the CAD model.

DIMENSIONAL METROLOGY OF AM PARTS

CT technologies are particularly useful to study the outcome of AM processes. The appeal of CT in dimensional metrology is evident in the recent growth of surveys in the field, e.g., see [38-41], where X-ray CT is considered a tool for dimensional quality control and geometrical tolerance verification of industrial components. One of the critical steps in evaluating AM performance of different systems would be to intentionally place simulated flaws of variated geometrical shapes inside the AM part designs. Then, after manufacturing, these features can be evaluated to identify the limits inherent to the AM process. Figure 6 presents one of these case studies that has been performed in collaboration with NIST (the National Institute of Standards and Technology) [37]. From the measurements of actual to nominal comparison, variations in dimensional geometry were detected for the AM parts with deviations up to ±0.1 mm with respect to reference geometry (nominal data). Material inclusions in the internal cavities were also detected as remnants from the AM process. The design of test parts for the evaluation of AM machines' dimensional performance has also been a subject of recent investigations. One example is the test object shown in Figure 7, which was first proposed by NIST [42]. Given its original geometry and large dimensions, such an object was designed so that the final AM part could be measured by CMM and surface texture instruments. However, it is not surprising to see researchers trying to use it for X-ray CT measurement, which, for a part with such high aspect ratios is challenging with metals.

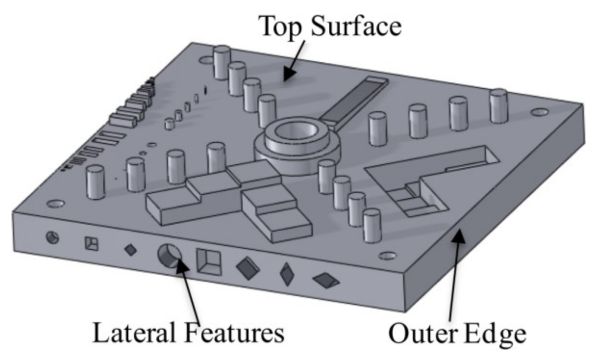

FIGURE 7. Solid model of a test part designed and proposed by NIST for AM performance evaluation and standardization purposes [42].

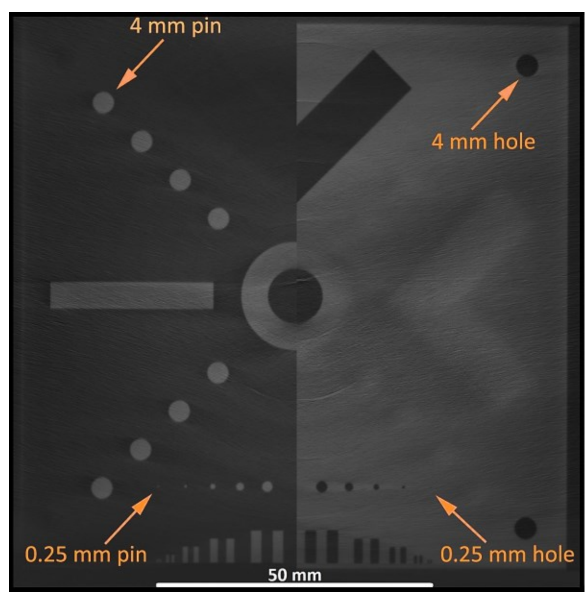

FIGURE 8. CT image of an AM built prototype of the NIST test part. The arrows in the figure point toward features and dimensions from nominal geometry of the NIST design (see Figure 7).

Metal parts often scatter X-rays, which may disrupt the CT reconstructions and produce unwanted artifacts in the data. Using a 2D fan beam of X-rays and a linear detector can reduce the scattering of X-rays and allow high quality CT data to be obtained. A highly penetrating beam, from a small X-ray source, is required for scanning highly absorbent materials such as nickel alloys. Figure 8 shows an image obtained by 2D fan beam X-ray CT of an AM prototype, made of Inconel 625, for the NIST test part. A Nikon XTH 450 system was used for CT data acquisition. For this sample, 420 kV was used with several millimeters of tin and copper filters to harden the polychromatic X-ray beam and reduce beam hardening artifacts. The image in Figure 8 was composed by joining two different halves of CT sectional slices, both extracted from equal separation distance from the 'top surface' but opposite in direction: the left half is 0.4 mm above from the 'top surface' and the left half is 0.4 mm below from the same surface. In future work, the authors plan to evaluate dimensional geometry assessments of the NIST test part based on three-dimensional CT data.

NEW PERSPECTIVES

One important focus of current research efforts in AM is to understand how process parameters influence defect formation in AM builds. Like any manufacturing process, there exists a large combination of input parameters (i.e., laser power, build speed, layer thickness, etc.) that can be chosen by the user before, during, and after the manufacturing process [43]. A complete, universal understanding of all of these inputs and how they affect the overall performance of a component is difficult to determine. Mathematical models based on empirical test data are one of the tools most commonly used for understanding the effects of AM process parameters [44]. To this end, statistical analysis of defect data can be used to determine the quality of a build. Given the correct design of experiments, optimization of process parameters can be performed to minimize the quantity and severity of internal defects. Given the vast amount of input parameters in many AM methods, this is not a speedy process. In many cases, it may require a large number of samples and large quantities for CT inspection. To shorten the time and effort in this process, some more recent research is leveraging advanced analytical methods such as artificial intelligence and machine learning algorithms [45]. These tools have the ability to analyze the complete defect data for a large set of samples to uncover hidden patterns and dependencies that may exist between the defects and input parameters, not apparent in other methods. There has also been growing interest in finding new in-situ techniques that can collect in-process data to feedback into the AM systems. This process would allow corrective actions by optimizing the parameters during the AM process. Integrating NDT techniques capable of sub-surface inspection during the AM process, such as X-ray CT, seem to be a challenging task, although some progress has been recently reported by using simple X-ray imaging [46-47] and X-ray phase contrast [48]. On the other hand, there has been growing interest in the possibility of using X-ray CT data for obtaining surface topography measurements of AM internal geometries without destroying or sectioning the part [49-52].

In general, a simple workflow optimization process to improve design and production of AM parts could be as follows: knowing the deviation from the intended form and dimensions of the design helps to quickly refine the model for an AM part (see Figure 9).

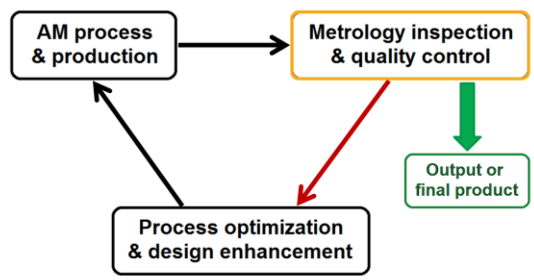

FIGURE 9. An iterative and cyclical optimization process for design and production in AM parts.

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